Experimental Methods & Reference Data for Menstrual Product Comparison

Specific, replicable protocols and published data mapped to each evaluation axis of this project. Sources: FDA guidance (Oct 2025), peer-reviewed studies, ISO standards.

AXIS 1: Safety — Bacterial Growth (Biology)

Metric: Colony Count (CFU/cm²) after 24h exposure

Recommended Method: Shake Flask Method (FDA-recommended, adapted from Schlievert & Blomster, 1983)

Why this method: The FDA's October 2025 draft guidance on menstrual product performance testing recommends three methods for microbiology assessment. The Shake Flask Method is the simplest and most replicable for a school lab setting.

Protocol

1. **Prepare bacterial culture:**
   - Use *Staphylococcus aureus* (ATCC strain, e.g., ATCC 6538 or ATCC 29213)
   - Inoculate in Brain Heart Infusion (BHI) broth
   - Incubate at 37°C until reaching ~1×10^8 CFU/mL (mid-log phase, ~$OD_{600} = 0.5$)
2. **Prepare test samples:**
   - Cut each product into standardized 1 cm² pieces (use sterile scissors)
   - For tampons: use a 1 cm section from the center
   - For pads: cut from the absorbent core area
   - Weigh each piece and record dry mass
3. **Inoculation:**
   - Add each sample to a sterile flask containing 50 mL BHI broth
   - Inoculate with *S. aureus* to a final concentration of 1×10^6 CFU/mL
   - Prepare control flasks: (a) broth only (negative control), (b) broth + bacteria without product (positive control)
4. **Incubation:**
   - Incubate at 37°C for 24 hours with shaking at 150 rpm
   - This simulates body temperature and provides aerobic conditions
5. **Enumeration (CFU counting):**
   - After 24h, perform serial dilutions (1:10 steps) of the broth
   - Plate 100 μL of appropriate dilutions onto BHI agar plates
   - Incubate plates at 37°C for 18–24 hours
   - Count colonies on plates with 30–300 colonies
   - Calculate: CFU/mL = (colonies counted) / (volume plated in mL × dilution factor)

    - Normalize to surface area: CFU/cm² = CFU/mL × total broth volume / sample area
6. **Expected results ranking (based on literature):**
    - Tampons (Ria, o.b.): Highest CFU — oxygen introduction + nutrient-rich environment promotes *S. aureus* growth
    - Commercial pads (Naturella, Always, Ria Ultra): Moderate CFU — synthetic materials may support surface growth
    - Organic pad (Jessa Cotton): Lower CFU — natural cotton, no synthetic polymers
    - Cloth pad (Jessa Cloth): Variable — depends on prior washing; clean cloth should be lowest

Alternative: Tampon Sac Method (FDA Method 1, adapted from Reiser et al., 1987)

- Place product in sterile permeable sac (e.g., dialysis tubing)
- Submerge in BHI agar (not broth) at 37°C for 18 hours
- Remove sac, homogenize contents, plate on BHI agar
- More realistic for internal products but requires dialysis tubing

Key Reference Data

Study	Finding	Relevance
FDA Guidance (2025)	Recommends minimum 1×10^6 CFU/mL *S. aureus* inoculum, 37°C, 24h incubation	Standard protocol
Schlievert & Blomster (1983)	Tampons can increase *S. aureus* growth 2–10× compared to broth-only control	Baseline for tampon risk
Reiser et al. (1987)	Tampon material composition affects bacterial growth rate; synthetic fibers > cotton	Explains product differences

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AXIS 2: Chemistry — pH Measurement

Metric: pH of product extract

Recommended Method: Aqueous Extraction + pH Meter

Protocol

1. **Prepare extraction solution:**
    - Use distilled/deionized water (or simulated vaginal fluid if available)
    - Simulated vaginal fluid recipe (Owen & Katz, 1999):
        - NaCl: 0.16 g/100 mL

   - KCl: 0.007 g/100 mL
   - $CaCl_2 \cdot 2H_2O$: 0.002 g/100 mL
   - Adjust to pH 4.2 with lactic acid
2. **Sample preparation:**
   - Cut each product into 1 cm^2 pieces
   - Weigh accurately (record dry mass in grams)
   - Place in sterile container
3. **Extraction:**
   - Add extraction solution at ratio of 10 mL per gram of product
   - Seal and incubate at 37°C for 24 hours (simulating body temperature and wear time)
   - Shake gently every 4 hours
4. **Measurement:**
   - Calibrate pH meter with standard buffers (pH 4.0, 7.0, 10.0)
   - Measure pH of each extract at room temperature
   - Record to 2 decimal places
   - Run each sample in triplicate
5. **Expected results:**
   - Healthy vaginal pH: 3.8–4.5 (acidic, protective)
   - Products closer to pH 4.0–4.5 are more compatible with vaginal environment
   - Products with pH > 6.0 may disrupt vaginal flora and increase infection risk
   - Commercial pads with chemical additives may shift pH away from natural range
   - Organic cotton should be closest to neutral (pH ~6–7 in water extract)

Key Reference Data

Study	Finding
Marcelis et al. (2021), *Emerging Contaminants*	Developed menstrual fluid simulant (MFS) with pH 4.2, osmolarity, and protein binding to assess chemical leaching from 15 MHPs
Owen & Katz (1999)	Vaginal fluid pH normally 3.8–4.5; rises during menstruation to ~5.0–6.0
FDA Guidance (2025)	Requires chemical identity disclosure for all components including fragrances and additives

AXIS 3: Absorption Capacity (Physics Exp 1)

Metric: Fluid held per gram of dry product (g/g)

FDA-Standard Method: Syngyna Testing (21 CFR 801.430)

This is the legally mandated method for tampon absorbency labeling in the US.

Syngyna Protocol (for tampons)

1. **Equipment:** Syngyna tester (available from lab suppliers)
2. **Test fluid:** 0.9% saline solution (0.9 g NaCl per 100 mL distilled water)
3. **Procedure:**
   - Weigh dry tampon (record mass)
   - Insert into Syngyna apparatus
   - Allow tampon to absorb saline from reservoir
   - After saturation, measure total saline absorbed
   - Absorbency (grams) = mass of absorbed saline
4. **FDA absorbency categories (21 CFR 801.430):**
   - Light: ≤ 6 g
   - Regular: 6–9 g
   - Super: 9–12 g
   - Super Plus: 12–15 g
   - Ultra: 15–18 g

Gravimetric Method (for pads — no FDA standard exists) Since the FDA does not regulate pad absorbency labeling, use this gravimetric method:

1. **Weigh dry product** (record mass in grams)
2. **Prepare test fluid:** 0.9% saline solution (or simulated menstrual fluid for more realism)
3. **Application:**
   - Place product on flat, impermeable surface
   - Slowly pour/pipe fluid onto the center of the product
   - Continue until fluid no longer absorbs (pooling or runoff occurs)
   - Wait 60 seconds for drainage
4. **Weigh saturated product**
5. **Calculate:**
   - Capacity (g/g) = (saturated mass – dry mass) / dry mass
6. **Repeat 3 times per product, report mean**

Published Reference Data (Real Blood — DeLoughery et al., 2024, *BMJ Sexual & Reproductive Health*) This is the first study to test menstrual products with actual human blood (not saline):

Product Type	Volume Held (mL)	Notes
Menstrual disc (Ziggy)	80	Highest capacity
Heavy pad (Brand C)	52	Advertised 10–20 mL
Tampon (Heavy, Brand A)	31	Advertised ~20 mL
Tampon (Heavy, Brand B)	34	Advertised ~20 mL
Tampon (Regular)	20	Standard
Menstrual cup (Size 2)	35	Largest cup size
Menstrual cup (Size 0)	22	Smallest cup size
Light day pad	4	Pantyliner
Period underwear (any size)	1–3	Lowest capacity

Key finding: Product labels significantly overstate capacity when tested with blood vs. saline. Heavy pads held 52 mL but were advertised at 10–20 mL capacity.

Why saline vs. blood matters (Scientific American, 2023)

- Blood is more viscous than saline □ absorbed at different rates
- Menstrual blood contains cells, platelets, proteins, vaginal secretions, and shed endometrial tissue
- Saline is homogeneous; menstrual blood viscosity changes hour-to-hour and person-to-person
- Research-quality human blood costs ~$100 per 10 mL; saline costs ~$45 per liter

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AXIS 4: Absorption Rate (Physics Exp 2)

Metric: Time to absorb 5 mL of simulated fluid (seconds)

Protocol

1. **Prepare test fluid:** 0.9% saline solution
2. **Set up:**
   - Place product on flat, impermeable surface
   - Position stopwatch
3. **Procedure:**
   - Using a pipette or burette, dispense exactly 5.0 mL of fluid onto the center of the product
   - Start timer the moment fluid contacts the surface

    - Stop timer when no visible fluid remains on the surface (all absorbed)
    - Record time in seconds
4. **Repeat 3 times per product, report mean**
5. **Lower time = faster absorption = better score**

Expected ranking (based on material properties):

- Commercial pads (Always Platinum, Naturella, Ria Ultra): Fast — designed with superabsorbent polymers (SAP)
- Tampons (Ria, o.b.): Moderate — absorption depends on fiber density and compression
- Organic pad (Jessa Cotton): Moderate-fast — natural cotton absorbs well but lacks SAP
- Cloth pad (Jessa Cloth): Slowest — natural fibers without chemical treatment

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AXIS 5: Environment — Biodegradation (ESS)

Metric: % Mass Loss after 14 days in soil

Recommended Method: Soil Burial Test (adapted from ISO 11721)

Protocol

1. **Prepare soil:**
    - Use garden soil or potting mix (sieve to remove large particles)
    - Adjust moisture to 40–60% water-holding capacity
    - Soil should be biologically active (not sterilized)
2. **Sample preparation:**
    - Cut each product into standardized pieces (e.g., 5 cm × 5 cm)
    - Dry in oven at 60°C for 24 hours
    - Weigh each piece accurately (initial mass, m_0)
    - Record mass to 0.01 g precision
3. **Burial:**
    - Place each sample in a mesh bag (nylon or cotton mesh, ~1 mm openings)
    - Bury bags 10–15 cm deep in soil
    - Keep soil moist (spray with distilled water as needed)
    - Maintain at room temperature (~20–25°C)
    - Label each bag clearly
4. **Retrieval:**
    - After 14 days, carefully excavate samples
    - Gently rinse off soil with distilled water

- Dry in oven at 60°C for 24 hours
- Weigh (final mass, m_1)

5. **Calculate:**
    - Mass loss (%) = $[(m_0 - m_1) / m_0] \times 100$
6. **Expected results (based on material composition):**
    - Jessa Cloth (cotton): Highest mass loss — natural fiber, readily biodegradable
    - Jessa Cotton (organic cotton): High mass loss — no synthetic components
    - Ria Tampon (cotton/rayon blend): Moderate mass loss — rayon is semi-synthetic but biodegradable
    - o.b. Tampon (cotton/rayon): Moderate mass loss — similar to Ria
    - Naturella / Always / Ria Ultra pads: Lowest mass loss — contain polyethylene, polypropylene, SAP (non-biodegradable plastics)

Published Reference Data (Brunsek et al., 2023)

Material	Mass Loss after Soil Burial	Notes
Hemp fiber	~40–60% (14 days)	Natural bast fiber, rapid biodegradation
Jute fiber	~35–55% (14 days)	Natural fiber, good biodegradation
Viscose (regenerated cellulose)	~30–50% (14 days)	Semi-synthetic but biodegradable
PLA (bio-based polymer)	~5–15% (14 days)	Very slow biodegradation in soil

Key Reference

Study	Finding
Paul et al. (2026), *Environmental Technology & Innovation*	Comprehensive review of biodegradable fibers for sanitary napkins; conventional pads are ~90% plastic by weight and persist in landfills for 500–800 years
Mirzaie et al. (2025), *Environ Sci Pollut Res*	LCA comparing bamboo pulp pads vs. conventional pads; conventional pads have significantly higher environmental impact across all categories
Brunsek et al. (2023)	ISO 11721 soil burial test methodology; mass loss is the standard metric for biodegradation assessment

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AXIS 6: Environment — CO_2 Footprint (ESS)

Metric: CO_2 equivalent per use (g CO_2e)

Recommended Method: Simplified Life Cycle Assessment (LCA)

Since a full LCA requires specialized software and databases, use this simplified approach:

Protocol

1. **Identify lifecycle stages:**
   - Raw material extraction
   - Manufacturing
   - Packaging
   - Transport
   - Use phase
   - Disposal
2. **Estimate CO_2e per product using published LCA data:**

Product Type	CO_2e per Use (g)	Source
Conventional disposable pad	5.5–14.0 g CO_2e	Mirzaie et al. (2025); Hait & Powers (2019)
Organic cotton pad	3.0–8.0 g CO_2e	Lower manufacturing impact, no synthetic polymers
Tampon (cotton/rayon)	2.5–6.0 g CO_2e	Smaller mass, but includes applicator (if applicable)
Tampon (no applicator, o.b.)	1.5–4.0 g CO_2e	No plastic applicator
Reusable cloth pad (per use)	0.1–0.5 g CO_2e	Amortized over 100+ uses; includes washing energy

3. **Calculation for reusable products:**
   - Total CO_2e (production) ÷ number of uses = CO_2e per use
   - Add washing CO_2e per cycle (~0.5–1.0 g CO_2e for machine wash)

Published LCA Data (Mirzaie et al., 2025) The study compared a bamboo pulp pad (Hempur) vs. a conventional pad:

Impact Category	Conventional Pad	Bamboo Pulp Pad
Global warming (kg CO_2e/kg product)	Higher	40–80% lower

Impact Category	Conventional Pad	Bamboo Pulp Pad
Water scarcity	Higher	Significantly lower
Abiotic element depletion	Higher	Significantly lower
Acidification	Higher	Lower

Key finding: Upstream operations (raw material + manufacturing) account for 40–80% of total environmental impact in every category.

Summary: All Methods at a Glance

Axis	Metric	Method	Standard/Reference
Safety	CFU/cm² after 24h	Shake Flask + serial dilution + plate count	FDA Guidance (2025), Schlievert & Blomster (1983)
Chemistry	pH of extract	Aqueous extraction at 37°C for 24h, pH meter	Marcelis et al. (2021), Owen & Katz (1999)
Capacity	g/g absorbed	Gravimetric (pads) / Syngyna (tampons)	21 CFR 801.430, DeLoughery et al. (2024)
Rate	seconds for 5 mL	Timed absorption on flat surface	Adapted from industry practice
Environment (mass loss)	% after 14 days	Soil burial test (ISO 11721 adapted)	Brunsek et al. (2023), Paul et al. (2026)
Environment (CO_2e)	g CO_2e per use	Simplified LCA using published data	Mirzaie et al. (2025), Hait & Powers (2019)

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